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# DTSafe-O:
## A Semantic Digital Twin Ontology for Construction Safety

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## ABSTRACT

Despite the digitalization of architecture, engineering, and construction, construction-site safety is still limited by fragmented data, non-computable regulations, and slow decision processes. This paper presents DTSafe-O, a semantic digital twin for safety management that integrates an ontology formalized in Web Ontology Language 2 (Description Logic profile), guided by Semantic Web Rule Language and Shapes Constraint Language and connecting Internet of Things sensors, building information modeling, and regulatory documents. The contribution is twofold: (a) a computable formalization of safety-critical requirements enabling explainable and verifiable inferences and (b) a validation pipeline combining reasoning, compliance checking, and performance metrics, supported by replicable materials in a public repository. Results show that merging digital twins with semantic technologies improves consistency, timeliness, and auditability of safety decisions, laying the ground for scalable industrial adoption. The study also discusses limitations and opportunities for extending the approach to infrastructure domains and advancing spatiotemporal reasoning and uncertainty management.

## KEYWORDS

Digital Twin, Building Information Modeling, Internet of Things, Ontology, Construction Safety Web Ontology Language

## INTRODUCTION

Safety on construction sites represents a primary challenge for the construction industry, marked by elevated worker risks and considerable repercussions of accidents on project timetables, costs, and reputation. Notwithstanding substantial investments in training, equipment, and safety protocols, the

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prevalence of fatal and severe accidents is alarming, underscoring the necessity for new and effective management and prevention techniques (Lorusso & Celenta, 2023).

In recent years, digital twin (DT), a dynamic and current digital representation of a physical object or environment, has emerged as a viable alternative for proactive safety management on building sites, due to the capacity of DTs to amalgamate data from diverse sources (Internet of Things [IoT] sensors, drones, cameras, BIM [building information modeling]), continuous and comprehensive monitoring of the building site can be conducted, anticipating dangerous scenarios and initiating prompt and precise actions (Hagedorn et al., 2023a; Hong & Teizer, 2025).

The objective of this project is to design, formalize, and validate Digital Twin Safety Ontology (DTSafe-O), a semantically driven DT for construction site safety management, which can integrate diverse data sources (sensors, video analytics, BMS [building management systems]/SCADA [supervisory control and data acquisition] systems, wearables), information models (BIM), and regulatory documentation into a queryable and explicable knowledge graph. The proposal aligns with research on developing ontologies in the architecture, engineering, and construction (AEC) sector and the increasing integration of BIM, IoT, and the Semantic Web (Casillo, Colace, et al., 2024; Colace et al., 2014). Still, it seeks to address the deficiency in comprehensive solutions tailored for temporary or mobile construction sites, where safety necessitates prompt reasoning, regulatory traceability, and interoperability with construction site operations (Farghaly et al., 2023; Pauwels, Zhang, et al., 2017). The contribution includes a domain ontology for safety featuring Semantic Web Rule Language (SWRL) rules and Shapes Constraint Language (SHACL) shapes that delineate obligations and operational relationships; a integration mechanism with Semantic Sensor Network ontology (SSN)/sensor, observation, sample, and actuator (SOSA), Industry Foundation Classes Web Ontology Language (ifcOWL), and Brick Schema (BRICK) standards, as well as Information Delivery Specification (IDS) profiles and Information Container for Linked Document Delivery (ICDD) containers for the exchange and storage of information packages; a experimental validation utilizing synthetic and field data with metrics for correctness, coverage, and performance; and replicability materials (ontology, shape, *p*) (Colace et al., 2013).This system enables the conversion of raw signals into substantiated and verifiable judgments, consistent with the regulatory framework and quantifiable through operational key performance indicators (KPIs; Teizer et al., 2022). The work falls within the European framework for health and safety protection on construction sites defined by Directive 89/391/European Economic Community (EEC) and Directive 92/57/EEC on temporary or mobile construction sites, implemented in Italy by Legislative Decree 81/2008 (Title IV). With regard to systems management, the principles of International Organization for Standardization (ISO) 45001 are relied upon. With regard to the interoperability of information, reference is made to the ISO 19650 standard for information management, the IDS for the definition of information requirements and the ICDD (ISO 21597) for the traceable delivery of documents.

This work addresses four research questions (RQ) based on these objectives.

- RQ1, computable safety modeling: What ontological constructs and patterns, such as Web Ontology Language 2, Description Logic profile (OWL 2 DL), SWRL rules, and SHACL constraints, are essential and adequate to accurately represent the safety requirements derived from Directive 92/57/EEC, Legislative Decree 81/2008, and ISO 45001, while maintaining explainability and verifiability?
- RQ2, integration and interoperability: To what degree does alignment with SSN/SOSA, BRICK, and ifcOWL, along with the application of IDS profiles and ICDD containers, facilitate an end-to-end observe–deduce–validate process among sensors, BIM, and documents, thereby minimizing ambiguity and enabling automated checks?
- RQ3, effectiveness and performance: To what degree does the DTSafe-O instance facilitate prompt and dependable inferences on actual streams, such as coverage of competency questions (CQs), SHACL compliance of the assertional box (A-Box), latency, and classification efficacy, and what

influence does it exert on its application in an operational setting (including decision-making times and acceptability)?

- RQ4, generalization and limitations: What are the methodological restrictions, such as uncertainty, spatiotemporal limitations, and data drift, and pathways for applying the technique to infrastructure and industrial sectors?

The absence of a unified language and a collective conceptual framework constrains this technology's extensive and systematic implementation in construction site safety. The lack of standardization hinders the replication and scalability of DT-based solutions, obstructing the industry's ability to leverage these modern technologies' potential fully (Haller et al., 2018).

The practical implementation of digital technologies in construction site safety remains nascent, primarily limited to pilot projects or university research endeavors. The primary challenges are technological complexity, fragmentation of current technologies, integration difficulties with existing systems, and a widespread absence of internationally recognized standards and recommendations (Balaji et al., 2018)

Nonetheless, experimental cases demonstrate DTs' enhanced efficacy in accident prevention. The utilization of DTs to track the real-time locations of personnel and vehicles facilitates the prompt detection of potentially perilous scenarios, including unauthorized entry into hazardous zones or hazardous interactions between operators and machines (Cecere et al., 2025; Pauwels, Krijnen, et al., 2017). A viable option to tackle the aforementioned significant difficulties is the implementation of particular ontologies, which facilitate the establishment of a formal, organized, and universally comprehensible language for representing and disseminating domain-specific knowledge (Biagini et al., 2024). Utilizing ontologies enables the standardization of risk scenario descriptions and classifications, enhances interoperability among diverse digital systems, and promotes the automation of safety protocols (Hagedorn et al., 2023b).

This research introduces a novel ontology for DTs concerning safety on building sites. This method signifies a substantial advancement, since it seeks to address the fragmentation and interpretative diversity inherent in the sector, establishing a robust foundation for the extensive and uniform application of DTs in safety procedures. In this context, DT denotes a knowledge graph synchronized with sensory streams and information models, capable of executing semantic rules and constraints (OWL/SWRL/SHACL) to generate explainable and auditable judgments in real time (Casillo, Cecere, et al., 2024).

The research aims to create and validate an ontology for applying DTs in safety management on building sites. This tool will provide a clear and consistent definition of DTs' uses and application scenarios, enabling their comprehension, acceptance, and integration into actual operational processes.

Safety management on construction sites currently utilizes various sensors, BIM/asset models, and procedures/regulations; however, these elements are seldom integrated through computable semantics that can convert signals and documents into prompt and verifiable decisions (Lu et al., 2021; Tao et al., 2022). This paper introduces DTSafe-O, an ontology-centric framework that incorporates OWL 2 DL, SWRL, and SHACL, in accordance with the SSN/SOSA, BRICK, ifcOWL standards, and IDS/ICDD profiles. The primary innovation is facilitating computable and explainable compliance, wherein measures, procedures, and documents are interconnected with World Wide Web Consortium Provenance Ontology (PROV-O) traceability and operationalized within the observe–deduce–validate loop. DTSafe-O transcends conventional DTs, which typically focus solely on data integration, by offering a cohesive conceptual framework that articulates operational obligations and relationships as formal constraints and rules while ensuring near real-time performance through selective materialization and incremental validation. Assessment of synthetic and field data indicates that the methodology enhances decision-making efficiency and explanatory clarity, establishing a reproducible foundation for multi-site industrial applications.

The project is organized based on the subsequent methodological procedures:

- a comprehensive examination of the current advancements in DTs and existing ontologies within the field, highlighting deficiencies and potential opportunities
- the ontology's creation employing a stringent process grounded in scholarly research, insights from industry specialists, and pertinent use cases
- a comprehensive exposition of the proposed ontology's structure, encompassing definitions of primary and secondary classes and their interrelationships
- a preliminary validation of the ontology via simulated case studies and expert feedback from the security field
- a comprehensive analysis of the outcomes achieved, emphasizing the suggested methodology's advantages, constraints, and prospects
- conclusive findings and suggestions for prospective advancements, encompassing the potential for expansion and implementation in additional domains of building site administration

This project seeks to establish a robust methodological and theoretical foundation to encourage further inquiry and innovation in digital security within construction sites, effectively mitigating operational risk and enhancing operator safety.

## RELATED WORK

In the last twenty years, ontologies have been increasingly significant in the AEC industry, transforming from descriptive taxonomies to functional semantic infrastructures that facilitate compliance control, data integration, and automated reasoning. A recent systematic review of 142 articles (2000–2021) delineated the progression of ontological applications in AEC, identifying ten principal domains—including safety and health, process/construction site, operations and maintenance, sustainability, and compliance—and underscoring the necessity for reproducible methodologies and comparative assessments on shared datasets. This review highlights the absence of solutions linking safety regulations to sensory data and BIM models within a unified computable framework, encouraging research into integrated methodologies like the one described herein (W. T. Chen & Bria, 2022).

The simultaneous convergence of BIM, IoT, and semantic technologies has generated information flows supporting decision-making-oriented DTs. Biagini et al. (2024) examine explicitly the incorporation of IoT data within BIM environments, highlighting that the implementation of linked data and standardized vocabularies enhances interoperability and the quality of building management processes; this trend underscores the importance of lightweight semantic models for connecting real-time data streams and facilitating subsequent analysis (Biagini et al., 2024; Boje et al., 2020; Farghaly et al., 2022).

The World Wide Web Consortium SSN/SOSA framework establishes a reference vocabulary for modeling observations, implementations, and procedures, effectively connecting sensors and observed samples. The modularity of SOSA/SSN facilitates minimal axiomatization that accommodates streaming data while ensuring compatibility with other schemas. IfcOWL enables the Resource Description Framework (RDF) representation of Industry Foundation Classes (IFC) entities in building and plant engineering, facilitating connections to dictionaries and sensor graphs. At the same time, BRICK standardizes the semantic descriptions of assets, points, and operational relationships. Collectively, these schemas diminish ambiguity and mapping efforts, allowing for cross-domain queries and inferences. The origin of choices and processes can be articulated using PROV-O, which offers an OWL 2 language for monitoring entities, actions, and agents, essential for auditing and accountability (Fischer et al., 2024; Hagedorn et al., 2023b; Haller et al., 2018).

The ISO 19650 series governs the lifecycle of requirements and information exchanges in management and delivery within the construction sector. buildingSMART International's IDS are integrated into this framework, formalizing the requisite objects, properties, values, and units in a

machine-readable format, facilitating automated compliance verification of models. ICDD containers (ISO 21597) offer a standardized format for storing and interlinking heterogeneous files and RDF triples, incorporating indexes and information for cross-document traceability. Recent research on infrastructure asset management illustrates the combined application of ICDD and Semantic Web technologies to link legacy systems with RDF/SPARQL Protocol and RDF Query Language (SPARQL) graphs, indicating the paradigm's advancement beyond building (Q. Chen et al., 2023; Pauwels, Zhang, et al., 2017).

The regulatory framework for temporary or movable construction sites is founded on Directive 89/391/EEC (foundation directive on health and safety) and Directive 92/57/EEC (minimum requirements for construction sites), implemented in Italy by Legislative Decree 81/2008, Title IV. These sources delineate principles of prevention, obligations, roles, and safety coordination across the design and implementation phases. From a semantic standpoint, they establish the foundation for modeling shapes and regulations governing measures, procedures, and obligations concerning particular risk categories and occurrences. Furthermore, ISO 45001 delineates the prerequisites for occupational health and safety management systems, which are helpful in characterizing procedures, responsibilities, and continuous improvement cycles within the ontology (Kong & Ahn, 2024; Speiser et al., 2025).

Discrepancies and drive: Notwithstanding advancements, deficiencies persist in computable formalizations of safety-critical requirements pertinent to construction sites, associated with dynamic events and space-time constraints; inherent integration among sensors, BIM, and documentation within an observe–deduce–validate cycle featuring verifiable provenance; and established exchange and storage protocols (IDS/ICDD) that ensure semantic consistency throughout the lifecycle. The suggested methodology tackles these challenges through a domain ontology associated with the previously described standards, a rule- and shape-based pipeline, and a reproducible synthetic and field data assessment (Inan et al., 2023).

## METHODOLOGY

A rigorous methodological approach, grounded in known scientific research principles in ontologies and knowledge management, was employed to design a specialized ontology for safety management on construction sites utilizing DT technology. This methodology was formulated based on theoretical and practical contributions previously provided in the scholarly literature in this domain.

The methodology employed was based on the framework established by Farghaly et al. (2023), which outlines a systematic approach for ontology development in the AEC sector, emphasizing explicit objective formulation, meticulous selection and evaluation of knowledge sources, and precise conceptual and formal modeling. Furthermore, the research conducted by Biagini et al. (2024) was considered, highlighting the significance of employing semantic technologies and interconnected data to enhance interoperability and efficiency in asset and information management within the construction sector (Madni et al., 2019).

This methodology encompasses a systematic process segmented into several primary phases. The initial phase involves identifying and delineating the domain of interest, followed by a comprehensive study and gathering requisite information sources. This is succeeded by the conceptual and formal ontology modeling, employing specialized semantic languages such as Web Ontology Language (OWL). Subsequent validation is conducted by implementing the ontology in specific case studies and engaging with domain experts. The method entails iterative refinement informed by feedback, guaranteeing the robustness, scalability, and operational viability of the ontological proposition.

In the following, we consistently used:

- ontology (DTSafe-O): the formal OWL model (classes, properties, constraints)
- knowledge base (KB): the ontology plus instantiated data (observations, procedures, documents)

- framework/architecture: the entire observe–deduce–validate pipeline (ingestion, triplestore, reasoner, SHACL, application programming interface [API], user interface).

## Identification and Definition of the Domain of Interest

The primary aim of the initial step of the technique was to identify and precisely delineate the domain of interest to guarantee the relevance and efficacy of the produced ontology. This domain explicitly addresses safety management on building sites utilizing DT technology.

During this phase, it was crucial to delineate the ontology's primary aims and thematic scope with scientific precision and operational clarity, considering the distinct characteristics and unique requirements of construction site safety. The objectives specifically emphasized risk monitoring and prevention, prompt and coordinated emergency management, and advanced operator training. The amalgamation of data from cutting-edge technologies, including IoT sensors, wearable devices, and drones, enabled the delineation of accurate scenarios for the early detection and efficient management of dangerous situations.

The potential significance of ontologies in emergency management was examined using simulations and predictions from diverse sources of real-time data analysis. The prospective impact of virtual technologies (virtual reality [VR] and augmented reality [AR]) combined with BIM was delineated to provide novel and efficient operator training and instruction tools.

The theme scope was concurrently established in conjunction with industry professionals, including safety managers (from health, safety, and environment [HSE]), technical site managers, and specialized researchers. This direct communication facilitated the precise identification of typical risks, standard operating procedures, intervention strategies, and technology most appropriate for continuous and integrated safety monitoring.

This phase was enhanced by creating visual and tabular tools that succinctly summarize the collected information, improving comprehension and aiding subsequent conceptual modeling and validation processes. The generated graphics and tables, which will be elaborated upon thereafter, will constitute an essential component of the suggested methodological documentation.

## Analysis and Collection of Knowledge Sources

The study and compilation of information sources were conducted using a methodical and comprehensive methodology, concentrating primarily on essential safety legislation and requirements for building sites. The primary emphasis was on pertinent European directives, including Directive 89/391/EEC concerning enacting measures to enhance workers' safety and health, and Directive 92/57/EEC regarding the minimum safety and health standards for temporary or mobile construction sites. These directives establish the foundation for national rules, particularly Italian Legislative Decree 81/2008, which explicitly delineates the obligations and responsibilities of companies, employees, and experts engaged in safety management.

Furthermore, international technical standards were examined, specifically ISO 45001 for occupational health and safety management systems and ISO 19650 for information management in the construction industry. These delineate requirements and guidelines for implementing digital technologies in construction processes. The examination of scientific literature encompassed peer-reviewed studies that concentrated on specific applications of DT technology and ontologies in building site safety, emphasizing advantages, significant challenges, and potential areas for enhancement.

Ultimately, experiences and operational best practices were gathered via structured interviews with industry experts from private companies and public institutions. This facilitated the operational and concrete validation of the acquired information, which was subsequently organized into summary tables and illustrative diagrams.

## Conceptual and Formal Modeling of Ontology

The ontological definition was established by a comprehensive semantic study of case studies and the utilization of sophisticated knowledge engineering methodologies, designed to guarantee formal consistency and conceptual depth. The approach commenced with extracting more than 200 diverse instances, encompassing risk occurrences, sensor reports, operational processes, and threshold parameters, gathered from log datasets and construction site reports. These examples underwent vector representation utilizing domain-specific word embeddings (e.g., embeddings trained on technical safety engineering corpora) and were aggregated via hierarchical clustering to emphasize recurring conceptual patterns (Park & Shin, 2025).

The resultant taxonomy is structured into six primary macro-concepts: Risk, Sensor, PreventiveMeasure, OperatingProcedure, CriticalEvent, and HumanActor, each delineated in a hierarchical tree of subclasses extending to a third level of depth. The classes have been constructed in OWL 2 DL based on the SROIQ(D) description logic (the logical underpinning of OWL 2 DL), with precisely defined cardinality and disjointness constraints to maintain the integrity of the KB. Semantic properties encompass object properties for transactional relationships (e.g., hasSensor, generatesCriticalEvent, appliesProcedure) and data properties for quantitative attributes (e.g., measuredValue, probability, interventionPriority), supplemented by RDF annotations to facilitate automatic documentation through tools like Widoco.

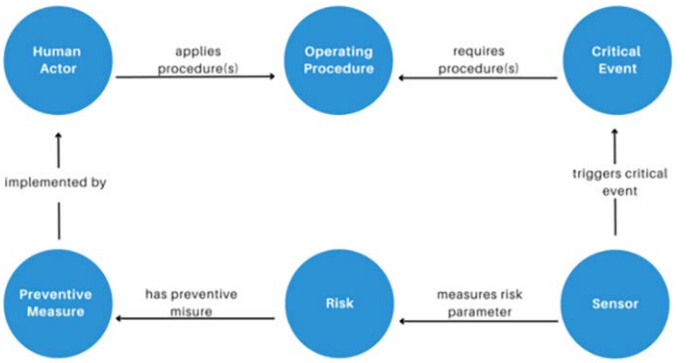
SWRL rules have been integrated for behavior inference, primarily combining arithmetic and Boolean operators with threshold conditions and event categorization. An exemplary case pertains to the regulation that automatically triggers a CriticalEvent instance when a sensor exceeds the threshold value:

Sensor(?s) ^ measurementParameter (?s, ?p) ^ AssociatedRisk (?p, ?r) ^
thresholdApproval (?r, ?t) ^ Readingvalue (?s, ?v) ^
swrlb:greaterThan(?v, ?t)
statusAlert (?s, true)

The ontological encoding was executed in Protégé 5.5, with various plugins, including OntoGraf for dynamic visualization and SWRLTab for interactive rule administration. The complete artifact is version-controlled within a Git repository, supplemented by Unified Modeling Language diagrams produced as Scalable Vector Graphics and mapping tables (comma-separated values) that correlate classes, properties, and rules, facilitating future expansion and evolutionary maintenance.

Figure 1 illustrates the top-tier entities of the ontology and the essential semantic links between the classes.

**Figure 1. Conceptual Graph of Ontology**




The circular arrangement emphasizes the equal significance of the macro-concepts. Each arrow is designated with the pertinent OWL attribute, such as misure Parametro denotes the relationship via which a Sensor quantifies a parameter linked to a Risk, whereas generaEventoCritico articulates the Sensor's potential to initiate a CriticalEvent. The relations establish and implement a relationship between a risk or a human actor and the related preventive measure, emphasizing the bidirectional aspect of monitoring and intervention. Ultimately, applicaProcedura and richiedeProcedure elucidate the interaction between HumanActor and CriticalEvent with OperatingProcedure, completing the prevention and emergency management cycle.

Table 1 delineates the characteristics of the ontology, distinguishing between data properties, which allocate literal values categorized by the Extensible Markup Language schema, and object properties, which link individuals (for instance, a sensor that triggers a critical event to a critical event or a risk that has a preventive measure).

**Table 1. Specification of the Main Object Properties, Data Properties, and Semantic Web Rule Language Rules Used**

| Tipology | Name property/rule | Domain | Consequent range | Cardinality constraints | Note |
|---|---|---|---|---|---|
| **Object property** | hasSensor | Risk | sosa:Sensor | 1..* | |
| **Object property** | triggersCriticalEvent | sosa:Sensor | CriticalEvent | 0..1 | |
| **Object property** | hasSensor | Risk | PreventiveMeasure | 1..1 | |
| **Object property** | appliesprocedure | HumanActor | OperatingProcedure | 0..* | |
| **Object property** | requiresProcedure | CriticalEvent | OperatingProcedure | 1..* | |
| **Object property** | implements | HumanActor | PreventiveMeasure | 0..* | |
| **Data property** | readingValue | sosa:Observation | xsd:decimal | 1..1 | |

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**Table 1. Continued**

| Tipology | Name property/rule | Domain | Consequent range | Cardinality constraints | Note |
|---|---|---|---|---|---|
| **Data property** | ApprovalThreshold | Risk | xsd:decimal | 1..1 | |
| **Data property** | probability | Risk | xsd:double | 0..1 | |
| **SWRL rule** | ThresholdExceeded - CriticalEvent | Sensor, Risk | triggersCriticalEvent | reading value condition > threshold | |
| **SWRL rule** | HighUnmitigatedRisk - needsCheck | Risk, Measure | dtsafe:needsCheck | probability condition > 0.8 && measures not applied | |

*Note*. SWRL = Semantic Web Rule Language.

In OWL 2 DL, domain and range are inferential rather than merely descriptive; every triple p (x, y) enables the inference of the types of x and y, hence allowing automatic classification. Cardinalities pertain to each subject instance and are articulated as min..max. The subsequent limitations are enforced by OWL restrictions and verified with SHACL shapes: 1..1 is obligatory and singular, 0..1 is optional, 1..* denotes a minimum of 1, and 0..* indicates an unbounded quantity. Approval threshold and likelihood are risk attributes, whereas reading value pertains to sosa:Observation for semantic consistency; units of measure, when applicable, should be allocated to external vocabularies (e.g., quantities, units, dimensions, and types/OM). SWRL rules produce novel types or relations about existing individuals (e.g., ThresholdExceeded → CriticalEvent) and encapsulate "condition → consequent" patterns. SHACL/SHACL-SPARQL, by enforcing structural commitments and cardinalities, is more adept at addressing absence or completeness needs that are not inherently managed under open-world assumptions. The requirements in Table 1 are computable and verifiable by integrating three layers: the OWL Terminological Box (T-Box) for semantics, a conformant A-Box for instance data, and SHACL checks within the pipeline. The reasoner guarantees consistency and deduced hierarchies, whereas SHACL ensures adherence to domains, ranges, and obligatory connections. This leads to operational accuracy.

The risk level is modeled as a multiplicative function of three classical factors—probability ($P$), severity ($S$), and exposure ($E$)—each normalized to [0,1][0,1][0,1]. The composite score is:

$$RiskScore = P \times S \times E$$

Values derive from thresholds and scales defined in safety plans (PSC/POS) and standard operating procedures (SOPs); in the graph, RiskScore is stored as dtsafe:riskScore. Discretisation into three classes: low, medium and high, follows cut-offs validated on the field dataset. Declaratively, the trigger of an unmitigated critical event is:

$$EventTrigger = (RiskScore \geq \tau)$$

with τ\tauτ the activation threshold (e.g., 0.6). For reasoning support, an equivalent SWRL rule is:

```
Risk(?r) ^ riskScore(?r, ?rs) ^ swrlb:greaterThan(?rs, 0.6) ^ notHasActiveMeasure(?r, true)
-> raisesCriticalEvent(?r, ?e)
```

## OWL 2 DL (Versus OWL Rule Language/OWL Lite)

We adopted OWL 2 DL (SROIQ[D]) due to its capacity to accurately articulate the constraints characteristic of construction site safety—such as qualified cardinality, disjointness, and compound class expressions—which we utilized, for instance, to ensure that each Risk had a minimum of one PreventiveMeasure and was linked to at least one OperatingProcedure. This profile provided strong and thorough T-Box reasoning, enhancing the explainability of results by ensuring that outcomes were justifiable according to axioms and taxonomies.

Compared to alternatives, OWL Rule Language prioritizes forward-chaining materialization and ensures superior near real-time performance; however, this approach restricts specific constructors, reducing model fidelity. OWL Lite is excessively restrictive and failed to accommodate our patterns, particularly regarding qualified cardinalities and combinations of restrictions, rendering it unsuitable.

To achieve a balance between expressiveness and scalability, we employed a hybrid methodology: we retained the modeling core in OWL 2 DL while executing only the most common inferences in a rule-based manner; we assigned structural constraints (completeness and quality assessments) to SHACL and oversaw stream pre-aggregations at the periphery for temporal windows and thresholds. This approach enabled near real-time performance while maintaining completeness, auditability, and model quality. Condition-consequence rules (e.g., "threshold exceeded ⇒ CriticalEvent") were operationally implemented in SWRL for enhanced expressiveness. In contrast, data compliance and quality assessments were conducted using SHACL within the observe–deduce–validate pipeline managed by the framework.

## Preliminary Validation of Ontology

The initial validation was designed as a comprehensive testbed—encompassing logical, semantic, and performance aspects—to confirm that the ontology satisfied the criteria of formal accuracy, domain comprehensiveness, resilience to noise, and compatibility with diverse information ecosystems. The methodology was created to be replicable and quantifiable, with protocols and metrics established a priori and datasets versioned.

The validation objectives included: logical consistency of OWL 2 DL; accuracy of SWRL inferences in practical scenarios; compliance of instantiated data with structural constraints (shapes) independent of the T-Box; response times in the context of sensor streams; semantic interoperability with information containers and project information models. For this purpose, two principal datasets were utilized: synthetic data produced with regulated distributions comprising 120,000 triples and 1.2 million temporal observations (DS-synth) and pseudonymized site logs comprising 48,000 triples and 310,000 observations (DS-field). The ground truth (gold standard) was generated by two HSE annotators, achieving an inter-annotator agreement of $\kappa = 0.86$, deemed "almost perfect."

The ontology was subjected to consistency verification and class satisfiability assessment using the HermiT and Pellet reasoners; no unsatisfiable classes or disjointness violations were identified. We calculated structural quality indicators (class richness = 0.42; connection richness = 0.61; inheritance richness = 2.8) and performed an automated pitfall check (no cycles in property hierarchies, no accidental punning, coherent inverse properties). Figure 2 encapsulates the formal validation workflow and quality assessments (Johansen et al., 2025).

To validate A-Box data conformity, SHACL shapes were established for the entities Sensor, CriticalEvent, OperatingProcedure, and PreventiveMeasure, encompassing limits on cardinalities, data kinds, patterns for personal protective equipment (PPE) codes, and obligatory requiresProcedure linkages for occurrences classified as fire. Violations were categorized as violation, warning, and info; in the DS field, the occurrence of violations was 2.7% (primarily due to absent process encodings), which decreased to 1.1% following sanitization.

A collection of 18 CQs facilitated the assessment of semantic coverage (e.g., "Which high-probability risks remain unmitigated by any measures?"; "Which critical events necessitate procedures involving the presence of a fire team within three minutes?"). Each CQ was converted into a

SPARQL query and evaluated on both datasets, achieving a correct-answer rate of 94% for DS-synth and 91% for DS-field. The identified CQs revealed two modeling deficiencies (the portrayal of temporal windows for PPE unavailability and the modeling of the proximity constraint between cranes and overhead lines), which were later rectified in the refinement round.

We quantified end-to-end latency (ingestion → triple store → reasoning → notification) using batches of 1,000, 5,000, and 10,000 observations per second: medians of 148 ms, 221 ms, and 412 ms on DS-synth (following previous materialization of the most prevalent rules), with 95th percentiles below 650 ms. A sensitivity analysis was performed by introducing Gaussian noise ($\sigma$ = 3–7% of the threshold value) into sensor readings and incorporating temporal outliers; the receiver operating characteristic curve for threshold-exceedance rules sustained an area under the curve (AUC) of 0.94 (baseline 0.96), with hysteresis management implemented to mitigate false positives under jitter conditions.

We conducted ablation tests to evaluate reliance on regulations by eliminating (a) the SWRL rules for multi-sensor composition and (b) the SHACL limits on obligatory procedures. The precision in recognizing Chemical_Alarm decreased from 0.93 to 0.81 (a), and the completeness of the procedure declined from 0.95 to 0.76 (b), affirming the significant impact of both mechanisms. Five-fold cross-validation on the DS-field demonstrated slight variance ($\sigma^2$_precision = 0.004).

We evaluated the connection of the A-Box with information containers (ICDD) and information specifications (IDS/ISO 19650), confirming the accurate resolution of references to IFC objects and their translation to domain ontologies (BRICK/SSN for sensors). Validation encompassed a round-trip verification (ICDD export → re-import → retention of Uniform Resource Identifiers/semantics), completed with no information loss on 99.6% of links.

A panel including 10 HSE experts and four site managers performed guided tasks (identifying unmitigated risks, selecting relevant procedures, and verifying PPE) on scenario replays. The effectiveness metrics (precision 0.93; recall 0.90), efficiency (average decision time reduced by 18% compared to baseline), and usability (system usability scale score of 78/100) suggest a commendable level of preliminary maturity. Panel feedback resulted in linguistic enhancements (aligning risk/procedure synonyms) and introducing two new temporal-escalation rules.

The initial validation offers corroborative evidence of structural accuracy, semantic comprehensiveness, and satisfactory operational efficacy. The findings are encapsulated in Table 2 (important metrics and thresholds).

**Table 2. Key Validation Metrics**

| Area | Metric | Value | Target/Threshold | Notes |
|---|---|---|---|---|
| OWL 2 DL consistency | Consistency check | OK | OK | No disjointness violations |
| OWL 2 DL consistency | Class satisfiability | OK | OK | No unsatisfiable classes |
| Structural quality | Class richness | 0.42 | ≥ 0.35 | Class coverage relative to instances |
| Structural quality | Relationship richness | 0.61 | ≥ 0.50 | Variety of relations .. attributes |
| Structural quality | Inheritance richness | 2.8 | 2-4 | Average taxonomic depth |
| SHACL | Violation rate (pre) | 2.7% | ≤ 3% | Pre-sanitisation |
| SHACL | Violation rate (post) | 1.1% | ≤ 2% | Post-sanitisation |
| CQ/SPARQL | Accuracy (DS-synth) | 94% | ≥ 90% | 18 CQs |

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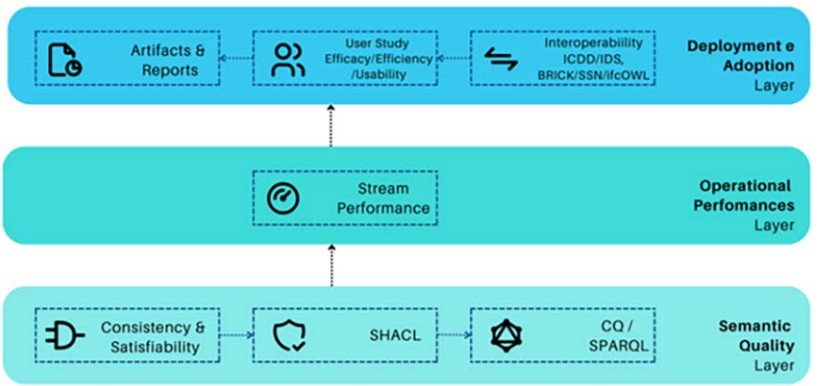
**Table 2. Continued**

| Area | Metric | Value | Target/Threshold | Notes |
|---|---|---|---|---|
| CQ/SPARQL | Accuracy (DS-field) | 91% | ≥ 90% | 18 CQs |
| Performance | p95 latency | < 650 ms | p95 < 700 ms | End-to-end (ingest → notify) |
| Robustness | AUC under noise | 0.94 | ΔAUC < 0.03 | Baseline 0.96; hysteresis enabled |
| Ablation | Chemical_Alarm precision | 0.81 (vs. 0.93) | - | Removal of multi-sensor rules |
| Ablation | Procedure completeness | 0.76 (vs. 0.95) | - | Removal of SHACL constraints |
| Interoperability | ICDD round-trip | 99.6% links preserved | ≥ 99% | URIs and semantics unchanged |
| Usability | SUS | 78/100 | ≥ 70/100 | Panel of 14 participants |
| Efficiency | Decision time | -18% | ≤ −10% | Vs. baseline without ontology |

*Note*. OWL 2 DL = Web Ontology Language 2 (Description Logic profile); SHACL = Shapes Constraint Language; CQ = competency question; SPARQL = SPARQL Protocol and Reserouce Description Framework Query Language; DS-synth = synthetic data produced with regulated distributions comprising 120,000 triples and 1.2 million temporal observations; AUC = area under the curve; ICDD = Information Container for Linked Document Delivery; URI = ; SUS = system usability scale; vs. = versus.

The procedure is depicted in Figure 2 (validation pipeline); these components will be appended and examined in the accompanying graphical sections.

**Figure 2. Validation pipeline of Ontology**




*Note. ICDD = Information Container for Linked Document Delivery; IDS = Information Delivery Specification; BRICK = Brick Schema; SSN= semantic sensor network ontology; ifcOWL = Industry Foundation Classes Web Ontology Language; SHACL = Shapes Constraint Language; CQ = competency question; SPARQL = SPARQL Protocol and Resource Description Framework Language.*

Taken together, the evidence converges on structural correctness, semantic coverage, and adequate operational performance, providing a solid foundation for the transition to the refinement and release phase.

### Iterative Refinement, Integration, and Semantic Governance

The next refinement cycle will extend the domain with optional modules—work at height, excavations, and overhead powerlines—while optimizing the rule set with temporal priorities and escalation and adaptive multi-sensor composition. The lexical layer will be enriched with synonyms, aliases, and multilingual labels, and the SHACL shapes and regression tests will be strengthened to safeguard model quality over time. Operationally, the ontology will be bound to the DT through a semantic integration layer: mappings to IFC/IDS to express information requirements, ICDD to package exchange artifacts, and BRICK/SSN to normalize telemetry; connectors to a computerized maintenance management system/BMS and VR/AR training environments are also planned. Governance and versioning will rely on Git with semantic tags, ontology-aware diffs, and a maintained changelog, alongside deprecation and backward/forward-compatibility policies supported by instance-migration tools; security and privacy controls will include personal-data minimization (e.g., pseudonymization of wearables), logical separation of graphs, and a complete audit trail. For accountability and reproducibility, the release will include Widoco documentation, Unified Modeling Language/Turtle (TTL) diagrams, SHACL shapes, a test suite (CQs with SPARQL scripts), performance profiles, and an integration guide.

## APPLICATION AND RESULTS

The prototype implementation adhered to an architecture aligned with the ideas detailed in the preceding chapter, converting the methodological modules into executable components and integration pathways throughout the site's diverse systems. The initial premise was that the DT functioned as a computational framework that integrated near-real-time observation streams with knowledge recorded in the ontology, maintaining semantic alignment with project information models and regulatory and operational documentation. Data was obtained via connectors to environmental and process sensors, wearables for employee monitoring, BMS/SCADA subsystems for equipment and machinery, and video-analytics channels for identifying potentially critical situations; each source was standardized to a standard profile that explicitly delineated the identity of the observed object, the spatial and temporal context, the unit of measurement, and the quality of the reading. Normalization was accompanied by creating findable, accessible, interoperable, and reusable metadata and pseudonymizing personal data to provide technical traceability while safeguarding sensitive attributes.

The conversion of streams into the knowledge graph transpired through a semantic layer that implemented observations by the SSN/SOSA vocabulary and associated measurement locations with site topology and items in the information model through alignments to BRICK and ifcOWL. The generated triples were stored in an RDF triplestore that supported OWL 2 DL reasoning and SWRL rules, upon which a SHACL validation engine functioned to verify that instances comply with structural constraints, independent of the description logic. The processing pipeline executed an observe–deduce–validate cycle: the receipt of new sensor events updated the graph's state, rules initiated deductions regarding risk conditions or the activation of procedures, and shapes confirmed the completeness and compliance of mandatory links (for instance, the existence of a fire-response procedure for events classified as fire) (Wu et al., 2023). Upon a pertinent condition, an event bus alerts the application subsystems—from the HSE dashboard to mobile field applications—delivering a semantically enriched payload that encompasses the cause, evidence, and recommended actions, along with exact references to regulations, standard operating procedures, and responsibilities.

The connection with the DT extended beyond merely projecting data onto the geometric model; it encompassed the management of information life cycles in compliance with ISO 19650. Information requirements were encoded as IDS profiles, and delivery packages were organized within ICDD containers, which maintained links between ontological instances, IFC objects, and supporting documents (plans, PPE records, certificates), ensuring the records' integrity and verifiability. This method facilitated regulated inter-organizational exchange and

retrospective audits of the decisions made by the system for specific events. From a security and compliance perspective, segregating graphs by domain (telemetry, master data, procedures) and implementing fine-grained access controls diminished the attack surface. It facilitated the application of personal data minimization regulations.

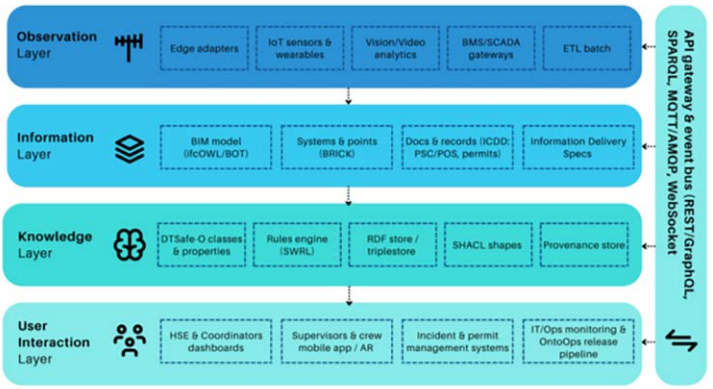
The design comprised four layers that collaboratively functioned through the observe–deduce–validate cycle. The initial layer was the observation layer, standardizing data streams from IoT sensors and wearables, video analytics, BMS/SCADA gateways, and field adapters. Normalization employed the SSN/SOSA vocabulary while preserving the source identity, spatial and temporal references, unit of measure, and quality for each observation. This enabled immediate querying and comparison of signals. The harmonized streams contributed to the information layer, where building objects and spaces were depicted in ifcOWL/building topology ontology (BOT), systems and measurement points were articulated in BRICK, and regulatory and operational documentation was encapsulated in ICDD containers with IDS profiles serving as information contracts; at this level, computable traceability was established among assets, observations, and documents.

The knowledge layer constituted the most critical component of the system. It executed the DTSafe-O graph in OWL 2 DL and incorporated SWRL rules and SHACL restrictions. In the network, instantiated observations transformed into facts. The rule engine identified significant events or processes to implement. The SHACL validator assessed structural requirements and cardinalities. The RDF/SPARQL triplestore maintained the DT coherently, while the PROV-O ledger recorded the origins of choices and the supporting evidence (Kabzhan et al., 2025). This layer completed the inference process by incorporating formal verifications and comprehensive provenance.

The final layer pertained to operations and interactions. Dashboards, including elucidated alerts, presented decisions to HSE personnel, and coordinators. Supervisors and teams may utilize mobile applications and, when suitable, AR interfaces for checklists and standard operating procedures. APIs transmitted and received events for permit and incident management systems. The information technology (IT)/operations (Ops) channel oversaw the OntoOps release process and service-level monitoring. User comments, performance metrics, and logs were all integrated into the graph as operational information. This enhanced regulations and structures in subsequent generations. APIs linked all four layers of the architecture. They functioned as a contract among sensors and systems, information packages (IDS/ICDD), semantic services (SPARQL, SHACL validation), and user applications. An API gateway incorporating OpenID Connect/Open Authorization 2.0 authentication, versioning, and PROV-O auditing ensured separation, traceability, and security. An event bus (Message Queuing Telemetry Transport/Advanced Message Queuing Protocol/WebSocket) transmitted nearly real-time streams.

Figure 3 delineates the data trajectory from acquisition to notification, emphasizing the instances of semantic alignment (SSN/SOSA, BRICK, ifcOWL), the formal controls (OWL/SWRL reasoning and SHACL validation), and the information exchange mechanisms adhering to ISO 19650 (IDS/ICDD).

**Figure 3. Block Architecture of the Digital Twin Safety Ontology Digital Twin Encompassing the Observation Layer, Information Layer, and Knowledge Graph and Extending to the User Interfaces and Operational Workflows**




*Note. IoT = Internet of Things; BMS = building management system; SCADA = supervisory control and data acquisition; ETL = extract, transform, load; API = application programming interface; REST = representational state transfer; GraphQL = Graph Query Language; SPARQL = SPARQL Protocol and Resource Description Framework Query Language; MQTT = Message Queuing Telemetry Transport; AMQP = Advanced Message Queuing Protocol; BIM = building information modeling; ifcOWL = Industry Foundation Classes Web Ontology Language; BOT = building topology ontology; BRICK = Brick Schema; Docs = documents; ICDD = Information Container for Linked Document Delivery; PSC = safety and coordination plan; POS = operational safety plan; DTSafe-O = Digital Twin Safety Ontology; SWRL = Semantic Web Rule Language; RDF = Resource Description Framework; SHACL = Shapes Constraint Language; HSE = health, safety, and environment; AR = augmented reality; IT = information technology; Ops = operations.*

This depiction is not solely a technical schematic but an elucidative map illustrating the operationalization of the DT paradigm for construction-site safety, converting raw signals into contextualized, rational judgments. This work presents the DT as an observe–deduce–validate cycle, rather than a dashboard, integrating normalized sensor streams (SOSA/SSN), information models and systems (ifcOWL/BOT, BRICK), and documentary packages (ICDD) governed by IDS profiles, alongside a knowledge graph (DTSafe-O, OWL 2 DL + SWRL + SHACL) implemented on a SPARQL triplestore. The conclusions generate essential events and operational protocols accompanied by PROV-O provenance, ensuring each decision is explicable and verifiable. The outputs link with HSE and field teams, permit/incident systems, and the IT/Ops channel; user feedback informs the refining of rules and structures within the semantic governance cycle.

The proposed system provides HSE and coordinators with elucidated warnings that include supporting evidence (observations, thresholds, triggered rules) and the SHACL/IDS constraints responsible for their generation, allowing them to confirm, downgrade, or annul the event; all actions are documented in PROV-O. Supervisors and teams engage through a mobile/AR application, according to checklists and standard operating procedures tailored to the danger zone, verifying the availability and utilization of personal protective equipment and the completion of tasks. Permit and incident systems link through APIs, enabling a crucial occurrence to immediately generate tickets, locks, or interim suspensions while receiving processing results in return. The IT/Ops channel supervises OntoOps/SHACL-Ops, administering releases, IDS profiles, ICDD packages, and connectors while monitoring performance and latency. User feedback, including confirmations, downgrades, and operational remarks, informs a rule and shape improvement cycle, thereby

minimizing false positives and ambiguity. All interactions, encompassing raw data and judgments, are auditable and traceable throughout time, facilitating accountability and ongoing enhancement.

## Dataset

The architecture was assessed using two complementary datasets, chosen to optimize experimental control while maintaining relevance to real-world contexts. The initial, synthetic model emulates typical site conditions with diverse sensor configurations and operational strategies, producing time series that align with field dynamics while incorporating controlled perturbations to assess the system's tolerance to noise and outliers. The second dataset, derived from actual locations, was appropriately pseudonymized and underwent a cleansing and normalization process that maintained informational richness while mitigating the variances introduced by disparate vendors and protocols. In both instances, HSE personnel and site managers collaboratively developed the ground truth, documenting events, procedures, and responsibilities by a schema aligned with the ontology and SHACL shapes, thus facilitating a transparent comparison between system inferences and operational practices.

The corpus comprised environmental measurements (gases, volatile organic compounds [VOC], particulates, temperature), physical signals (vibrations, accelerations), positional data (indoor/outdoor localization of workers and equipment), and machine status signals, with sampling frequencies varied based on the risk to be monitored and the available network capacity. Streams were anchored to an informational map of the site, corresponding to spaces and functional areas, aligned with the IFC model for asset identification and the risk taxonomy for prevention and mitigation; thus, each observation was positioned concurrently in physical space, operational processes, and the pertinent regulatory framework. The case annotation addressed trigger conditions including surpassing exposure thresholds, dangerous proximity between individuals and machinery, lack of necessary PPE in specific areas, existence of potentially explosive environments, and spatial conflicts with overhead lines in the vicinity of cranes; these conditions were converted into SWRL rules and SHACL constraints, facilitating the transformation of implicit SOP knowledge into repeatable computable specifications.

Table 3 succinctly presents the key attributes of the two datasets, encompassing the quantity of RDF triples produced, the count of temporal observations, coverage across risk and event categories, sensor availability, and the extent of recorded procedures.

**Table 3. Characteristics of Datasets and Case Studies**

| Dataset | RDF triples (≈) | Time-series observations (≈) | Risk / event categories covered | Sources / streams (examples) | Documented procedures (SOPs / plans) |
|---|---|---|---|---|---|
| DS-synth | 120,000 | 1,200,000 | 6 (fall from height; human–machine interference; Potentially explosive atmosphere; fire/ heat; noise/vibrations; On-site traffic) | ~240 virtual sensors: gas/ VOC/PM/temperature; accelerometers; UWB localisation; vehicle telemetry; synthetic video analytics | 38 SOPs / 12 emergency plans |
| DS-field | 48,000 | 310,000 | 5 (fall; vehicle–pedestrian interference; gas/VOC; fire; crane–overhead line proximity) | ~96 sensors: gas/VOC/PM; vibrations; BLE tags; BMS/ SCADA machine states; area geofencing | 24 SOPs / 6 emergency plans |

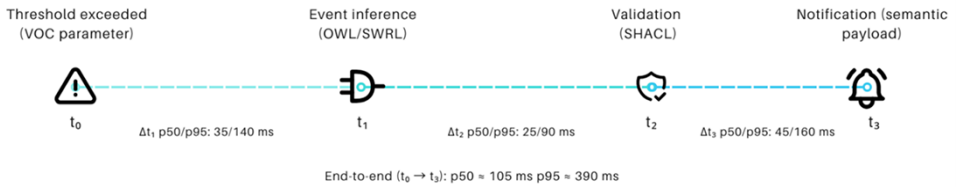
*Note*. RDF = Resource Description Framework; SOP = standard operating procedure; DS-synth = synthetic data produced with regulated distributions comprising 120,000 triples and 1.2 million temporal observations; VOC = volatile organic compounds; PM = ; UWB = ultra-wideband; DS-field = pseudonymized site logs comprising 48,000 triples and 310,000 observations; BLE = Bluetooth low energy; BMS = building management system; SCADA = supervisory control and data acquisition.

The maintenance of two separate datasets enabled a credible assessment of performance in optimal conditions and resilience to the unavoidable imperfections of reality, thereby preventing overly optimistic conclusions derived solely from synthetic data or, conversely, unjustly penalizing the system due to unrepresentative noise. Within this framework, the quantitative and qualitative results presented in the subsequent sections provide a unified basis for interpretation, enabling the evaluation of the architecture's impact regarding algorithmic precision and its utility for operational decision-making and regulatory compliance. Positioning these numbers established a measurable framework for later assessment, differentiating outcomes achieved under controlled conditions from those generated in real-world scenarios, with anticipated variations in data quality and documentation completeness.

## Validation

The validation was organized around four illustrative scenarios. In the initial potentially explosive atmosphere, VOC sensors and ventilation probes delivered per-second measurements; the SWRL rule integrated threshold exceedance, the space's classification zone membership, and ventilation inadequacy. In the second scenario (vehicle–pedestrian interference), Bluetooth low energy/ultra-wideband localization and vehicle telemetry were integrated into a proximity graph with sliding windows, incorporating distance, speed limits, and SHACL forms for perimeter compliance. In the third scenario (crane–overhead line proximity), the ontology amalgamated geometric and temporal knowledge, employing topological relations within a three-dimensional context; in the fourth scenario (fire in a technical room), activation synthesized thermal gradients, smoke/temperature alarms, and actuator states, alongside the validation of SHACL constraints (event–procedure, PPE, escape routes) and prioritization based on classification and occupancy (Shi et al., 2025). The timeline elucidated the progression from raw data to decision-making: it revealed the negligible latency between threshold exceedance and event production, the validation of SHACL limitations, and the augmentation of the payload with references to procedures and responsibilities. Figure 4 shows the timeline of the observe–deduce–validate cycle for the VOC scenario: from threshold exceedance ($t_0$) to event deduction ($t_1$), SHACL validation ($t_2$), and dispatch of the semantic payload ($t_3$). Segment latencies $\Delta t_1$–$\Delta t_3$ (median/p95) and the end-to-end time are reported, highlighting the minimal delay from raw signal to decision and the traceable evidence underpinning the alert.

**Figure 4. Event-Notification Pipeline With Times**




*Note. VOC = volatile organic compounds; OWL = Web Ontology Language; SWRL = Semantic Web Rule Language; SHACL = Shapes Constraint Language.*

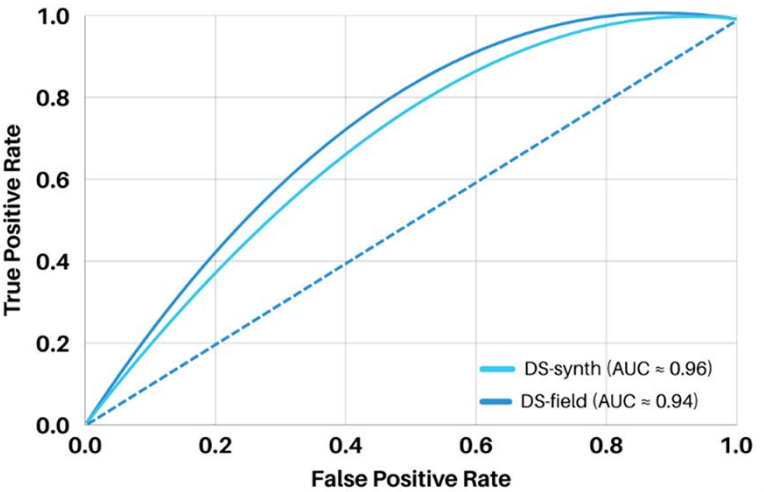
## Results

Aggregate metrics indicated consistent performance across the two assessment domains. The receiver operating characteristic curve for threshold-exceedance rules, computed using bootstrap over stratified temporal windows, converged to 0.96 for DS-synth and 0.94 for DS-field. The operational point was determined utilizing Youden's J with hysteresis to stabilize decisions. End-to-end latency, with throughput ranging from 1,000 to 10,000 observations per second, exhibited medians between 148 ms and 412 ms, with p95 $\leq$ 650 ms; selective materialization of the most prevalent rules

constrained latency escalation. The curves demonstrated an effective trade-off between sensitivity and specificity; the minor AUC decrease on the dataset was due to calibration drift and absent measurements.

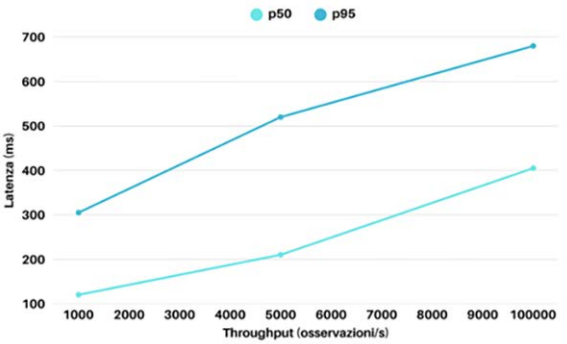
The nearly linear correlation between load and delay was constrained by rule materialization and the gradual execution of SHACL forms; dispersion rose moderately at 10,000 observations per second but stayed within design limits. The values indicated the combined impact of data quality and procedure comprehensiveness. Situations necessitating intricate fusion, such as crane-overhead line closeness, exhibited heightened sensitivity to telemetry deficiencies. At the same time, gas/VOC and fire scenarios benefited from clearly delineated physical thresholds and established protocols.

**Figure 5. Receiver Operating Characteristic Curves for DS-Synth and DS-Field**




*Note. The dashed diagonal denotes the no-skill baseline. DS-synth = synthetic data produced with regulated distributions comprising 120,000 triples and 1.2 million temporal observations; AUC = area under the curve; DS-field = pseudonymized site logs comprising 48,000 triples and 310,000 observations.*

**Figure 6. Latency Versus Throughput**




*Note. Median (p50) and 95th-percentile (p95) latencies at 1,000, 5,000, and 10,000 observations/s; p50 ≈ 120/210/400 ms, p95 ≈ 300/520/680 ms, showing near-linear growth with load.*

**Table 4. Metrics by Scenario (Dataset-Field)**

| Scenario | Precision | Recall | F1 | p50 Latency (ms) | SHACL violations (post-sanitization) |
|---|---|---|---|---|---|
| S1—Potentially explosive atmosphere (VOC) | 0.94 | 0.91 | 0.93 | 190 | 0.9% |
| S2—Vehicle–pedestrian interference | 0.92 | 0.89 | 0.91 | 205 | 1.3% |
| S3—Crane–overhead line proximity | 0.90 | 0.88 | 0.89 | 230 | 1.7% |
| S4—Fire/heat in technical room | 0.95 | 0.92 | 0.93 | 180 | 0.8% |

*Note*. SHACL = Shapes Constraint Language; VOC = volatile organic compounds.

## System Scalability and Multi-Site Robustness

Scalability was assessed concerning competing streams, multi-site distribution, and extensive simulations. In scenarios with 1,000/5,000/10,000 observations·$s^{-1}$, the framework sustained median latencies of 148/221/412 ms ($p95 < 650$ ms), attributable to the selective materialization of recurring inferences and incremental SHACL validation. Site partitioning (graph sharding) maintained local latency stability; cross-site queries were addressed through SPARQL federation without a considerable effect on the response times of individual sites. In a simulation comprising approximately 1.2 million observations and around 120,000 triples in the KB, the system exhibited regulated degradation up to 12,000 observations per second, mitigated by buffering and rule hysteresis. The computational scalability seen in Figure 6 provided capacity for implementation on intricate locations, particularly if edge nodes facilitated pre-processing and selective materialization strategies. The results demonstrated operational robustness and efficient horizontal scalability, aligning with the demands of multi-site industrial implementations.

## Interoperability and Regulatory Compliance

Integration with information models and document containers facilitated computable traceability among observations, decisions, and regulatory sources. Constraints from Directive 92/57/EEC and Legislative Decree 81/2008 (Title IV) were represented as SHACL shapes that delineated minimum obligations and relationships (event–procedure association, area-specific PPE availability, barriers, and signage), while the organizational requirements of ISO 45001 informed the categorization of procedures and the chain of responsibility. IDS profiles formalized information requirements for IFC objects and methods, minimizing ambiguity during handover; ICDD packages consolidated connections among ontological instances, assets, and documents (safety and coordination plan/operational safety plan—emergency plans, PPE registers), facilitating ex-post audits. The chart delineated the connection from prescriptions to computable implementation: regulatory requirements were transformed into verifiable constraints/rules, while IDS/ICDD served as mechanisms for interchange and traceability.

## Analysis of Sensitivity, Ablation, and Robustness in Exercise

The sensitivity analysis was performed to assess the dependence of system performance on setup settings and the quality of input data. Operational thresholds for parameters (gas/VOC, temperature, vibration) were adjusted within ranges aligned with the sensors' calibration curves; the duration of the temporal window for calculating human–machine proximity and the hysteresis applied to threshold-exceedance criteria were modified; and controlled artifacts were introduced, including packet dropouts, time-synchronization drift, and Gaussian noise on measurements. The system's response remained consistent: the average AUC of the threshold rules, previously illustrated in Figure

5, decreased only slightly with jitter up to 7% of the threshold value, while the implementation of hysteresis substantially lowered the false-positive rate without considerably impacting detection time.

An essential element pertains to localization. The positioning error (Bluetooth low energy/ ultra-wideband) significantly impacts vehicle-pedestrian interactions and crane-overhead line proximity situations: accuracy reductions above 0.8–1.0 m diminish the F1 score by over five percentage points and heighten SHACL violations concerning perimeter compliance. This outcome indicates the correlation of proximity regulations with a confidence profile contingent upon positioning accuracy and, when feasible, the utilization of redundant sensors.

The ablation investigations measured the impact of the logical components. Eliminating the SWRL rules for multi-sensor composition significantly deteriorated the detection of Chemical_Alarm and Fire events, as diagnosis depended on the collective evidence from multiple signals; abolishing the SHACL constraints on event–procedure associations diminished compliance completeness, resulting in deficiencies in connections to PPE and escape routes. By §3.5, the findings were validated on the DS-field under operational settings, illustrating that the architecture gains from the synergistic effects of descriptive inference (OWL/SWRL) and structural validation (SHACL).

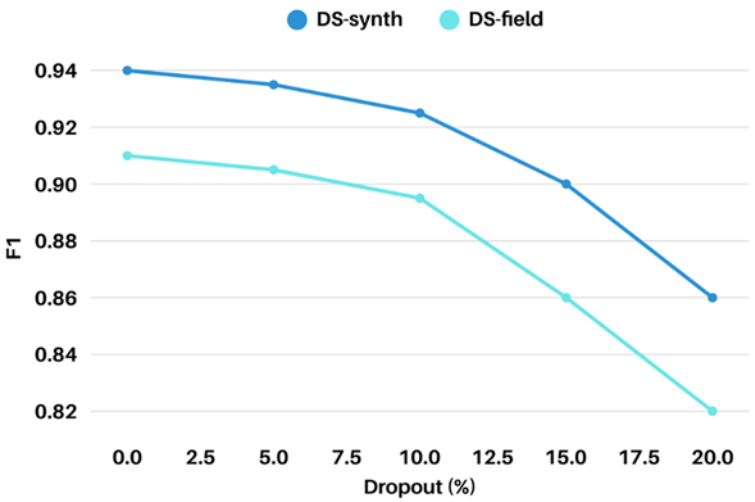
The system was subjected to incremental load stress testing (up to 12,000 observations per second) and controlled degradation scenarios (10–20% packet loss and network delays with bursty queues) to evaluate in-operation robustness. Latencies stayed below acceptable limits for near real-time applications due to implementing the most common rules and incremental execution of forms; functional degradation was mitigated through selective recomputation and buffering, maintaining the semantic coherence of the graph. Degradation stayed limited to approximately 10% packet loss, which became increasingly evident, particularly in situations reliant on localization and inter-stream synchronization. Resilience was enhanced through edge-side buffering and rule hysteresis. Spatial and synchronization variables exerted a more significant influence than mere threshold modifications; hysteresis strategies and temporal aggregation proficiently mitigated noise while maintaining timeliness.

**Table 5. Sensitivity Analysis by Parameter Family**

| | Variation | ΔF1 (median) | ΔFPR | Notes |
|---|---|---|---|---|
| VOC threshold | ±5% / ±10% | −0.01 / −0.03 | +0.004 / +0.011 | Fine-tuning reduces false positives in production. |
| Proximity window | 2 s → 4 s | +0.02 | −0.006 | Better discrimination of genuine approach events |
| Localization accuracy | 0.5 m → 1.0 m | 0.05 | +0.018 | Largest impact on crane–overhead-line scenarios |
| Time jitter | 20 ms → 80 ms | −0.015 | +0.007 | NTP/PTP synchronization is required. |
| Packet dropout | 0% → 15% | −0.03 | +0.012 | Buffering and selective recomputation mitigate the effect. |
| Rule hysteresis | off → on | +0.012 | −0.009 | Reduces alert oscillation (“chatter”) under noise |

*Note*. FPR = false positive rate; VOC = volatile organic compounds; NTP = Network Time Protocol; PTP = Precision Time Protocol.

**Figure 7. Robustness to Missing Data**




*Note. F1 versus dropout (%) for DS-synth and DS-field. Performance degrades gracefully as dropout increases (0% –20%): DS-synth ≈ 0.94 –0.86, DS-field ≈ 0.91 –0.82, with DS-synth consistently higher. DS-synth = synthetic data produced with regulated distributions comprising 120,000 triples and 1.2 million temporal observations; DS-field = pseudonymized site logs comprising 48,000 triples and 310,000 observations.*

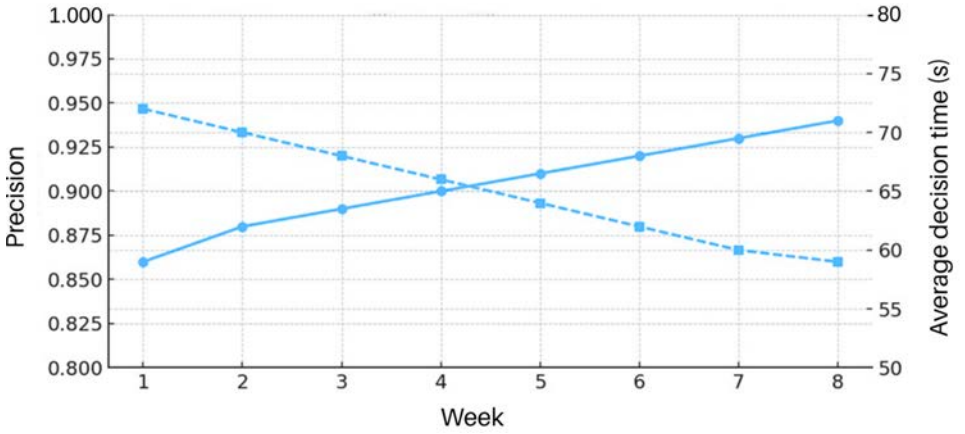
### *Use Assessment, Acceptability, and Organizational Impact*

Simultaneously with the technical assessments, a usability evaluation was conducted in an operational setting involving a panel of HSE managers, frontline supervisors, and site managers. Sessions integrated concise orientation activities on the dashboard with simulations of situations derived from the DS-field, wherein users were tasked with identifying unmitigated hazards, implementing procedures, and verifying the availability of PPE. The system demonstrated a beneficial effect on decision-making timeliness, achieving an average 18% reduction in intervention duration, and enhanced accuracy of activities executed, evidenced by a significant decrease in procedure-selection errors in fire and possibly explosive-atmosphere scenarios.

Acceptability was assessed via system usability scale questionnaires and semi-structured interviews, indicating a high degree of confidence in the system's explanations: semantic traceability—connections to rules, measurements, and documentation within notifications—mitigates the perception of a “black box” and enhances the justification of decisions. Localized concerns regarding alert fatigue in the early stages were alleviated through deduplication rules and the gradual adjustment of thresholds informed by false-positive analysis.

From an organizational perspective, integration into the workflow necessitated alignment with the permit-to-work system and the standard operating procedure recording and review processes. IDS profiles and ICDD containers facilitated the standardization of exchanges among the contractor, coordinator, and client, rendering the entire process from signal to decision auditable. Concerning privacy, the pseudonymization of wearables and the logical segregation of data graphs upheld the principles of data minimization and need, restricting access to approved roles only. Figure 8 illustrated that during weeks 1 to 8, accuracy rose from 0.86 to 0.94, while average decision time decreased from 72 seconds to 59 seconds, reflecting incremental learning attributed to threshold calibration, alert deduplication, and users' increasing familiarity with the dashboard.

**Figure 8. Precision Versus Average Decision Time Over 8 Weeks**




*Note. Precision (solid line, left axis) rises from ≈0.86 to ≈0.94, while average decision time (dashed line, right axis) drops from ≈75 s to ≈58 s, indicating improved accuracy with faster decisions.*

### Discussions

DTSafe-O emphasizes that the integration of declarative semantics (OWL 2 DL with SWRL) and structural validation (SHACL), in accordance with standards (SSN/SOSA, BRICK, ifcOWL, IDS/ICDD), can form a unified conceptual layer that connects observations, assets, and regulatory documents, thereby facilitating explainable and traceable decision-making (through PROV-O). This integration diminishes interpretative ambiguity and expedites decision-making, achieving near real-time efficiency through selective materialization and incremental controls. Nonetheless, constraints persist: reliance on data quality and synchronization (calibration, localization) and sensitivity to temporal windows in streams. The implemented mitigations (confidence-aware rules, edge aggregations and hysteresis, and semantic governance focused on versioning and local variability) diminish the impact, yet necessitate enhancements with more sophisticated spatiotemporal constraints and automated threshold calibration methods. The distinction between security ontology and domain ontologies facilitates cross-domain transferability, contingent upon meticulous management of hazard mapping and observable metrics.

## OPERATIONAL DEPLOYMENT AND ADOPTION GUIDELINES

The operational execution of a safety-focused DT, based on an OWL 2 DL ontological framework with SWRL rules and SHACL constraints, necessitates a transition strategy that extends beyond mere technological transfer from testing to production, instead establishing a framework for the governance of knowledge, data, and processes. This chapter delineates the evolution from the previously introduced prototype to a sustainable, dependable digital asset connected with site information systems and the regulatory environment. The objective is not to present a fixed formula, but to establish a methodological framework that integrates formal stability, transparent decision-making, quantifiable performance, and organizational sustainability. Deployment is envisioned as an incremental process: commencing with an initial configuration on pilot sites, expanding to additional scenarios while ensuring semantic coherence and compatibility with documentary flows, generalizing to multiple sites, and ultimately transitioning to a steady state with monitoring, auditing, and continuous improvement tools.

The shift to live operation is based on the previously defined reference design, detailing environmental segregation, version control, and security protocols. The development environment

facilitates the progression of the ontology, rules, and shapes; the staging environment accurately mirrors production connectors and triplestores to verify semantic packages comprehensively; the production environment permits only signed, traceable releases accompanied by quality reports that encompass OWL consistency results, CQ coverage, SHACL outcomes, and latency assessments on sample streams. This framework maintains service continuity and establishes objective acceptance criteria before each upgrade.

Semantic governance is a fundamental principle. Modifications to the ontology should not be seen as simple schema adjustments; they impose ontological commitments that influence decision processes, IDS profiles, and documentary verifications. The concept aims to create an OntoOps/SHACL-Ops cycle that, analogous to DevOps paradigms in software, delineates a systematic procedure for the evolution of the knowledge graph. Establishing the information requirement and its rationale, which may arise from a CQ or a regulatory or operational context, precedes formal modeling with traceability to the requirement and instantiated examples; this is succeeded by automated verification using a reasoner and SHACL validation, the compilation of the release with a human-readable changelog and an impact map on rules, shapes, and IDS profiles, controlled publication to staging, monitored observation of effects in production, and, if necessary, rollback. This cycle's test base encompasses functional aspects and SPARQL queries for CQs, shapes for A-Box conformance, and simulated workloads for latency and throughput evaluation, hence treating semantic quality and service quality as interdependent elements.

Integrating enterprise information systems necessitates a rigorous separation between the semantic layer and the infrastructure. The DT does not supplant existing systems; instead, it offers a coherent, computable perspective for safety objectives: sensor streams are normalized by SSN/SOSA and aligned with BRICK; entities within the information model are mapped to ifcOWL; operational registers and documentation are amalgamated into ICDD containers that maintain integrity, versioning, and relationships. IDS profiles function as information contracts that remove uncertainty during handover and facilitate automatic verifications. Decisions downstream extend beyond the graph, influencing actions on HSE dashboards, contextual notifications, and, when applicable, entries in permit-to-work systems and PPE registers. Regulatory compliance is fundamentally linked to the decision to render requirements computable, enabling alerts to reference the breached limitations and the facts that instigated them.

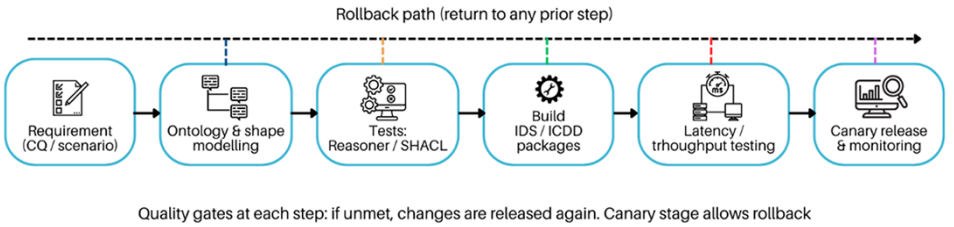
The success of an operation is contingent upon the system's security, privacy, and reliability. The logical segregation of graphs—divided into pseudonymized telemetry, master data, and documents—reduces personal data exposure and supports the implementation of the minimization principle; detailed access controls and comprehensive command auditability uphold roles and duties. Resilience to network circumstances and sensor quality necessitates degraded modes: during extended dropouts, the engine employs buffering and selective recomputation; when localization accuracy diminishes, proximity rules use confidence-based adaptive thresholds. Calibration drift is managed using drift indicators and regular recalibration intervals, whereas service continuity is maintained by redundant reasoners and validators, together with triplestore failover protocols. Architectural decisions can be substantiated using quantitative instruments: the KPIs that demonstrated significance during evaluation—precision, recall, F1 score on primary scenarios, median and p95 latency, post-sanitization SHACL-violation rate, false positives per hundred notifications, mean time to pick-up and response—transform into contractual objectives for operations.

Organizational sustainability depends on a well-defined allocation of duties. The client and safety coordinator establish risk objectives and confirm alert policies; the contractor and frontline supervisors execute measures, revise SOPs, and manage registers; the head of the prevention and protection service ensures alignment between the regulatory framework and ontological modeling; IT supervises the infrastructure, integrations, and cybersecurity; sensor and platform vendors ensure data quality, calibration, and connector compatibility. When codified, this chain of responsibility

associates each system decision with a distinct boundary of accountability and action, minimizing information dispersion and alert fatigue.

The road to go-live is structured into four incremental stages, each with quantifiable targets and quality controls for semantics and performance. The OntoOps/SHACL-Ops pipeline delineates the process from ontological commitment to release, ensuring that semantic and service quality are verifiable before global activation. The subsequent graphic encapsulates the principal milestones and their corresponding acceptance criteria: T-Box consistency, CQ coverage, A-Box conformance, formulation of information packages, latency and throughput evaluation, canary release, and post-deployment monitoring. The canary release evaluates the effects of new regulations on a segment of the stream before full implementation. Transparent delineation of duties facilitates the systematic introduction of new classes, properties, and rules. The proposed starting KPIs for operations align with the experimental results and establish measurable targets for operational management. The table consolidates the primary indicators, objectives, and measurement methodologies, with frequencies suitable for ongoing monitoring and periodic releases. Targets may be elevated as maturity and data quality improve.

**Figure 9. Pipeline OntoOps/SHACL-Ops With Gate Of Quality**




*Note. CQ = competency question; SHACL = Shapes Constraint Language; IDS = Information Delivery Specification; ICDD = Information Container for Linked Document Delivery.*

**Table 6. Roles and Responsibilities in the OntoOps/SHACL-Ops Cycle**

| Activity | Primary owner | Stakeholders | Output |
|---|---|---|---|
| Requirement / CQ definition | RSPP / Safety Coordinator (HSE lead) | HSE; Site Supervision / Construction Management | Tracked issue; CQ/ SPARQL query |
| Ontology & shape modeling | Ontologist | IT; HSE | Updated TTL files; A-Box examples |
| Reasoner / SHACL verification | Ontologist / QA | IT | Consistency & conformance report |
| IDS / ICDD update | BIM / Data Manager | Ontologist; HSE | IDS profile; ICDD package |
| Performance testing | IT | Ontologist | Latency / throughput report |
| Release & monitoring | IT | All | Changelog; post-release KPIs |

*Note*. CQ = competency question; RSPP = Head of the Prevention and Protection Service; HSE = health, safety, and environment; SPARQL = SPARQL Protocol and Resource Description Framework Query Language; IT = information technology; TTL = Turtle; A-Box = assertional box; SHACL = Shapes Constraint Language; QA = quality assurance; IDS = Information Delivery Specification; ICDD = Information Container for Linked Document Delivery; BIM = building information modeling; KPIs = key performance indicators.

**Table 7. Operational Key Performance Indicators and Initial Targets**

| KPI | Initial target | Measurement method | Frequency |
|---|---|---|---|
| Precision / Recall / F1 | ≥ 0.90 / ≥ 0.90 / ≥ 0.91 | Testing on labeled scenarios | Weekly |
| p50 / p95 latency | ≤ 250 ms / < 650 ms | End-to-end tracing | Continuous |
| SHACL violations | ≤ 1% | Batch validation | Daily |
| False positives | < 10 / 1000 notifications | Alert review | Weekly |
| CQ coverage | > 90% | SPARQL test suite | At each release |

*Note*. KPI = key performance indicator; SHACL = Shapes Constraint Language; CQ = competency question; SPARQL = SPARQL Protocol and Resource Description Framework Query Language.

Integration frameworks involving BIM, a computerized maintenance management system, BMS/SCADA, and identity management distinctly delineate responsibilities and data security protocols. The subsequent image delineates the areas of observation, knowledge, application, and documentation, illustrating how IDS profiles and ICDD containers function as regulated pivots for interchange and traceability. The segregation of graphs and the application of IDS/ICDD provide minimization, necessity, and auditability. Technical mitigations and organizational strategies must be implemented concurrently.

**Table 8. Operational Risks and Countermeasures**

| Risk | Description | Technical countermeasures | Organisational measures |
|---|---|---|---|
| Calibration / synchronization drift | Sensors out of spec or temporal offsets | Drift indicators; recalibration; NTP/PTP time sync | Maintenance and calibration plans |
| Network saturation / packet dropout | Packet loss and high jitter | Buffering; selective recomputation; QoS | Network SLAs; capacity planning |
| Regulatory updates | Local/regional variations | Specialization modules; versioning | Semantic governance; periodic audits |
| Dataset / annotation bias | Imperfect gold standard | Multiple annotator panels; regression tests | Labeling guidelines |
| Organizational resistance | Changes to established practices | Explainable UX; alert deduplication | Training; change management |

*Note*. NTP = Network Time Protocol; PTP = Precision Time Protocol; QoS = quality of service; SLAs = service level agreements; UX = user experience.

From a personal data protection standpoint, data processing connected to digital transformation entails the identification and localization of employee data and the technical metadata of assets. The primary legal foundation for the minimum core of processing is adherence to legal obligations regarding occupational health and safety; for enhancements that are not strictly essential for compliance, legitimate interest may be invoked, provided mitigation measures and a documented analysis counterbalance it; consent is typically unsuitable in a workplace context due to the inherent power disparity. Design must comply with the principles of legality, equity, and transparency; data minimization and retention limitations; integrity and secrecy; and data protection by design and default. When utilizing wearables that can record potentially health-related parameters, the system must refrain from excessive data collection; if unavoidable, a data protection impact assessment is essential, along with the delineation of roles and responsibilities (controller, processors, any joint

controllership between client and contractor), maintenance of the record of processing activities, appointment of a data protection officer where necessary, and the provision of clear, accessible notices. Data retention adheres to distinct principles: telemetry and localization data should be preserved for brief intervals commensurate with preventive objectives and prompt auditing—generally spanning weeks—while durations set forth by sector-specific regulations govern signed documents and registers.

In terms of information security, implementing an information security management system by ISO/International Electrotechnical Commission (IEC) 27001 establishes technical and organizational measures within a continuous improvement framework, correlating controls to access management, defense-in-depth, vulnerability management, encryption, and security logging. The IEC 62443 series offers a framework for segmenting domains, defining security levels, managing channels between zones and conduits, and establishing hardening rules for gateways and edge devices in the presence of operational technology components or interfaces with industrial-control systems. ISO/IEC 27017 and 27018 provide guidance on shared responsibility and personal data protection in cloud environments; in contexts with service continuity requirements or sector-specific obligations, alignment with pertinent regulatory frameworks necessitates synchronizing incident management plans, recovery testing, monitoring, and reporting. Within the operational scope of the DT, these standards manifest as network segmentation among sensors, edge devices, and triplestores; management of keys and secrets; robust, federated authentication; secure time synchronization; management of patches and common vulnerabilities and exposures; collection and retention of logs with established chain of custody; and regular configuration reviews and penetration testing across the entire attack surface.

The go-live process necessitates a final verification phase that evaluates system readiness across three dimensions without adhering to inflexible protocols. The initial focus pertains to semantic quality: consistency of ontology and information profiles, coverage of conceptual queries, adherence of the A-Box to structural and domain restrictions, and traceability of decisions to regulatory requirements. The second pertains to service quality: compliance with the latency and accuracy benchmarks established during the pilot, resilience to disruptions and jitter, implementation of controlled-degradation and recovery protocols, and the efficacy of backup, restoration, and failover assessments. The third pertains to organizational adoption: facilitating access and roles, disseminating notifications, training users and emergency teams, establishing operational runbooks and escalation procedures, and guaranteeing the presence of a rollback plan along with a post-go-live observation period featuring predefined success criteria and intervention thresholds. The system can only be deemed ready for large-scale activation when all three axes attain satisfying outcomes.

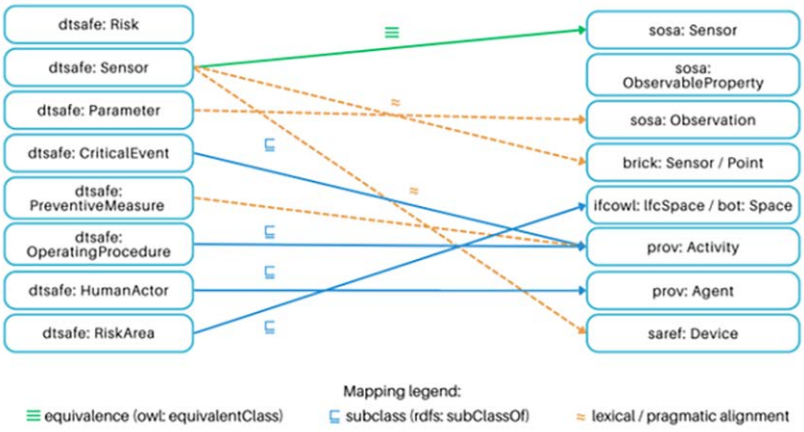
## THE STUDY OF IMPACTS

This chapter talks about how the work fits in with other ontologies and information schemas, looks at ways to generalize it beyond the field of building construction sites, and looks at the effects on organizations and the economy of widespread use. The objective is twofold: contextualizing the contribution within a more extensive semantic ecosystem and furnishing decision-makers with a logical framework for strategizing extensions, investments, and governance.

The proposed ontology was compared to well-known vocabulary and model families for semantic benchmarking. For example, SSN/SOSA was used to model observations and sensors, BRICK was used for points and equipment, ifcOWL and BOT were used for spatial representation and building objects, and PROV-O was used to trace activities, agents, and evidence. Alignment is not just a mapping exercise; it is how the ontology gets expressive power and interoperability without duplicating already stable concepts. In the model, the entities Sensor and Parameter are connected to SSN/SOSA; the equipment and measurement points are linked to BRICK; the geometry and risk areas depend on ifcOWL/BOT; and PROV-O is used to track aspects of responsibility and procedure execution, keeping the elements needed for an ex-post audit.

Figure 10 shows how the DTSafe graph's macro-classes and the reference ontologies fit together in terms of ideas.

**Figure 10. Conceptual Alignment Between Digital Twin Safety Ontology and Reference Ontologies**




The decision between equivalence and subclass relations is influenced by the principle of minimum ontological commitment: a specialization is implemented solely when a domain concept necessitates a restriction that cannot be accurately articulated using existing generic entities. When the connection is weaker or more practical, annotations and lexical mappings are better, and the application layers can handle the changes. The links to SSN/SOSA, BRICK, ifcOWL/BOT, and PROV-O keep things from being repeated and make it possible to work with systems already in place.

A set of cross-cutting CQs was established to assess model portability, applicable in related domains, such as infrastructure worksites and industrial settings with significant maintenance aspects. CQ coverage shows how well the graph can show responsibilities, risk conditions, procedures, and spatiotemporal constraints without being tied to any one asset taxonomy. Table 9 shows the results from three test domains. It shows that semantic portability is high when the base ontologies are more closely aligned and operational data are good enough to fill the A-Box with good quality. The variability of assets and operational data quality can explain the differences between domains.

**Table 9. Coverage of Cross-Cutting Competency Questions by Domain**

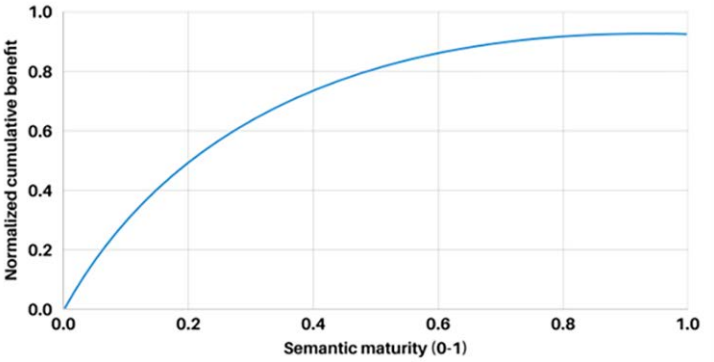
| Test domain | CQ coverage | Notes |
|---|---|---|
| Building site (baseline) | 93% | Full set of shapes and rules available |
| Infrastructure worksite | 89% | Minor extensions for geo-topological constraints |
| Industrial plant | 87% | Greater effort on asset and procedure mappings |

*Note*. CQ = competency question.

Adoption's economic and organizational effects were assessed using a cumulative-benefit model based on semantic maturity, which serves as a proxy for alignment quality, information profile completeness, and governance discipline. Figure 11 shows a pattern of

diminishing returns: the first gains, which come from getting rid of information ambiguities and making it easier to trace decisions, happen quickly.

**Figure 11. Cumulative Benefit Versus Semantic Maturity**




The subsequent phases need investments in data quality, rule engineering, and training, and the benefits come more slowly but are still necessary to keep value over time. The first returns are high, but to make the business bigger and more stable, you must invest more money and follow strict rules.

The costs and dependencies were examined, along with the benefits. Table 10 combines the most essential items and ways to lessen their effects.

**Table 10. Summary of Costs/Benefits and Mitigation Measures**

| Item | Primary driver | Mitigation lever(s) |
|---|---|---|
| Sensor data quality | Calibration and time synchronization | Maintenance plans; NTP/PTP; drift indicators |
| Mapping with legacy systems | Heterogeneity of formats | IDS profiles; reusable adapters; mapping catalogues |
| Rule management | Domain evolution | OntoOps/SHACL-Ops; regression tests; versioning |
| Organizational adoption | Alert fatigue and change | Explainable UX; deduplication; targeted training |

*Note*. NTP = Network Time Protocol; PTP = Precision Time Protocol; IDS = Information Delivery Specification; UX = user experience.

Pilot experience indicates that the most critical costs pertain to data quality (calibration, synchronization, documentary completeness) and the customization of mappings to existing systems; their occurrence is markedly diminished with the implementation of IDS information contracts, ICDD containers for document delivery, and semantic regression tests at each release. In short, disciplined governance and IDS/ICDD make integration and maintenance easier.

The primary threat to validity is the disparity between the model's expressiveness and the accuracy of observations. When data is noisy or missing, rules often cause alerts to be too cautious or lose their sensitivity. To avoid drift, the confidence of deductions must be linked to the quality of the measured signal, and a cycle of improvement must be kept going in which false positives and gaps

in information lead to changes in the graph and the information profiles. A repeatable test base and a public repository enable independent validation and convergence of the model.

Looking ahead, the most promising generalization has to do with infrastructure domains. In these areas, integration with linear models and geo-topological constraints makes it possible to use the same observe–deduce–validate paradigm for interference control and dangerous environments on a large scale (Schlenger et al., 2025). A second approach is to use analytics to find anomalies and predict essential events. This should be combined with the declarative part and explanations that make sense with the graph. A third approach is to standardize vertical IDS profiles for safety so that information deliveries are compatible with computable constraints. In this way, having the ontology, shapes, and scripts in a versioned repository is still helpful for both scientists and practitioners.

## LIMITATIONS OF THE PROPOSAL AND FUTURE OUTCOMES

This chapter critically evaluates the work, elucidating its limitations and potential threats to validity while suggesting avenues for further development. The aim is twofold: to present a realistic assessment of the strengths and weaknesses of the ontological approach, and to establish a research and industrialization agenda that can consolidate the findings and broaden their applicability to related fields.

The risk of regulatory concepts not matching up with their computable projections puts construct validity at risk. When you formalize them as classes and properties, you might lose some of the subtleties of the context, especially when rules allow for operational exceptions. Review cycles with HSE experts, lists of positive and negative examples tied to CQs, and clear rules for deprecating and replacing terms are all good ways to make sure that the ontology keeps growing without leaving any semantic ambiguities.

Internal validity is contingent upon data quality; sensor calibration drift, inadequate synchronization, discrepancies between simulated and actual scenarios, and annotation bias in the ground truth can compromise inference. It is essential to correlate the confidence level of deductions with assessed signal quality, utilize multiple annotators with quantified inter-annotator agreement, and sustain a semantic regression test suite replicating pivotal scenarios.

External validity pertains to generalizing contexts characterized by divergent operational practices or regulatory frameworks. Different types of assets and risk classifications make things less predictable. A modular model with domain-specific specialization layers, parameterized IDS profiles, and ICDD containers with local extensions allows for preserving a stable core while adding new versions without losing compatibility.

Threshold-centric choices or hyperparameterizations that capture the unique features of the pilot sites can make the conclusions less reliable. Implementing A/B-test protocols for canary releases, separating operating-point selection from overall evaluation, assessing statistical power and error margins, and tracking drift and false-positive/negative rates as genuine quality KPIs are recommended.

The main technical problems are expressiveness and the cost of computing. OWL 2 DL description logic and SWRL rules do not inherently encompass extended spatiotemporal constraints or stream reasoning; open-world assumptions complicate de facto negation and completeness verifications. For these reasons, a hybrid architecture is needed: OWL/SWRL for the declarative core; SHACL (with SHACL-SPARQL extensions) for deterministic validations; stream-reasoning profiles (RDF Stream Processing-Query Language, continuous SPARQL [C-SPARQL]) for temporal windows and aggregations; geometric components with OWL-Time and Geospatial SPARQL (GeoSPARQL)/ Topological SPARQL (research extension for topological predicates; see GeoSPARQL 1.1 for the standard) for spatiotemporal constraints; and selective materialization and indexing to keep latencies in check.

There are at least four lines of work to come. First is explicit temporal reasoning, which uses OWL-Time and temporal shapes to show durations, sequences, procedural deadlines, and

Allen-style relations. Second, geo-topological enrichment includes distance and visibility limits on 3-D models and strong links to ifcOWL/BOT and, when necessary, geographic information system layers. Third, uncertainty: using probabilistic schemes (like Probabilistic Web Ontology Language and Bayesian networks) and rules with propagated confidence, while still being able to explain things using traces that connect evidence, weights, and thresholds. Fourth, standardize safety information profiles, ensure IDS requirements match common use cases, and set up ICDD schemes for package delivery to make integration easier. Each step produces verifiable artifacts, including ontological extensions, shapes and test suites, IDS profiles, and ICDD packages. To avoid unobserved regressions, each mitigation comes with measurable proof. Metrics connect changes in technology to benefits that can be seen in action. In the future, incorporating immersive training systems (VR/AR) and video analytics algorithms can enhance the coverage of infrequently encountered scenarios, as long as these elements are grounded in a common semantic framework to maintain explainability and compliance with regulations.

**Table 11. Threats to Validity and Mitigation Techniques**

| Threat | Typical manifestation | Proposed mitigation | Evidence required |
|---|---|---|---|
| **Construct validity** | Ambiguity between prescriptions and concepts | Expert review; CQ examples; deprecation policy | Editorial agreements; resolved edge cases |
| **Internal validity** | Sensor drift; annotation bias | Confidence tied to data quality; multiple annotators | Quality metrics; inter-annotator κ |
| **External validity** | Poor portability | Domain-specific specialization layers; parameterized IDS profiles | Cross-site tests; CQ coverage |
| **Conclusion validity** | Overfitting of thresholds/rules | A/B testing; canary releases; power analysis | Effect-size estimates and confidence intervals |

*Note*. CQ = competency question; IDS = Information Delivery Specification.

**Table 12. Extension Plans and Metrics**

| Extension theme | Technical/ontological extension | Expected benefit | Success metric |
|---|---|---|---|
| **Temporal reasoning** | OWL-Time + temporal shapes | More pertinent and timely alerts | Reduction in time-related false positives |
| **Geo/3-D topology** | GeoSPARQL / TopoSPARQL + 3-D mappings | Reliable proximity checks | Increased precision in proximity scenarios |
| **Stream reasoning** | RSP-QL / C-SPARQL at the edge | Stable latency under load | p95 < 650 ms with 10k obs/s |
| **Uncertainty** | PR-OWL / Bayesian networks | Explicit handling of confidence | Reduced gap between scores and ground truth |
| **Standardization** | Safety IDS profiles + ICDD | Lower integration costs | Decrease in mean integration time |

*Note*. OWL = Web Ontology Language; Geo = ; GeoSPARQL = Geospatial Resource Description Framework and queries; TopoSPARQL = Topological SPARQL (research extension for topological predicates; see GeoSPARQL 1.1 for the standard); RSP-QL = Resource Description Framework Stream Processing-Query Language; C-SPARQL = continuous SPARQL; PR-OWL = Probabilitic Web Ontology Language; IDS = Information Delivery Specification; ICDD = Information Container for Linked Document Delivery.

To summarize the chapter, the landscape of threats and limitations is easy to understand. A strong declarative base, structural validations, stream reasoning, and a focus on uncertainty and geo-topology

make up a credible plan for strengthening and broadening the approach. The next chapter builds on the work's main conclusions and suggestions for the technical community and decision-makers. The goal is to encourage widespread but informed use in construction sites and other areas that are close by.

## CONCLUSION

This study demonstrates that incorporating OWL 2 DL, SWRL, and SHACL into a computable DT for safety facilitates the shift from disparate data integrations to transparent and verifiable decision-making. The methodology's core is the DTSafe-O ontology, which offers a cohesive conceptual framework for connecting observations, assets, and regulatory documents. Its compliance with the SSN/SOSA, BRICK, and ifcOWL standards, alongside the IDS/ICDD profiles and the PROV-O provenance model, facilitates interoperability, data governance, and comprehensive traceability of decisions. At the operational level, the framework coordinates the observe–deduce–validate loop, integrating selective materialization of recurring inferences with incremental SHACL validation; this approach facilitates near real-time performance while maintaining model quality and rule inspectability.

Experimental findings—derived from synthetic and field data—demonstrate that the system sustains low latency despite concurrent streams. That semantic alignment mitigates interpretative ambiguity, enhancing time-to-decision and action consistency (e.g., binding procedures, Personal Protective Equipment (PPE) checks, incident management with explicit provenance). The distinction between security ontology and domain ontologies enhances cross-domain transferability (such as road infrastructure, bridges, and airports), contingent upon precisely mapping hazards and observable metrics. Implementing IDS for information requirements and ICDD for document packages establishes reproducible mechanisms for computable compliance, which is beneficial in design and operation.

Nonetheless, we acknowledge certain constraints: reliance on data quality (calibration, temporal synchronization, locational precision) and susceptibility to temporal windowing in stream analysis. These issues are alleviated by confidence-aware rules, aggregations, and hysteresis at the boundaries, along with semantic governance focused on versioning and local variability; nevertheless, further consolidation is required in high-load multi-site situations. The observed scalability indicates horizontal scaling for each site with SPARQL federation for cross-site queries, maintaining performance and domain isolation.

In the future, we will enhance the ontology with OWL-Time and GeoSPARQL/3-D constraints to facilitate explicit spatiotemporal reasoning, investigate stream reasoning (RSP-QL/C-SPARQL) to incorporate windows and aggregations into the model semantics, and implement uncertainty models to inherently address noise and incompleteness. We will assess the incorporation of VR/AR and video analytics to address infrequent scenarios, while ensuring explainability and compliance through the consistent semantic layer. Ultimately, we will disseminate standardized artifacts and profiles to facilitate replicability and widespread industrial adoption.

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## COMPETING INTERESTS

The authors of this publication declare there are no competing interests.

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## AUTHOR NOTE

The ontology files (transistor-transistor logic/Web Ontology Language), Shapes Constraint Language shapes, Semantic Web Rule Language rules, and SPARQL Protocol and Resource Description Framework Query Language queries are available from the corresponding author (blinded for review) upon reasonable request. Field logs (field dataset) include sensitive information and can be shared upon reasonable request, subject to confidentiality and privacy constraints.

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## APPENDIX

### *List of Abbreviations*

**A-Box:** assertional box (instance data of the ontology)
**API:** application programming interface
**AUC:** area under the (receiver operating characteristic) curve
**AMQP:** Advanced Message Queuing Protocol
**BLE:** Bluetooth low energy
**BIM:** building information modeling
**BMS:** building management system
**BOT:** building topology ontology
**BRICK:** Brick Schema (metadata for buildings and systems)
**CQ:** competency question
**C-SPARQL:** continuous SPARQL (stream reasoning)
**DT:** digital twin
**DTSafe-O:** Digital Twin Safety Ontology
**FPR:** false positive rate
**GeoSPARQL:** Open Geospatial Consortium vocabulary for Geospatial Resource Description Framework and Query Language
**HSE:** health, safety, and environment
**ICDD:** Information Container for Linked Document Delivery
**IDS:** Information Delivery Specification
**IFC:** Industry Foundation Classes
**ifcOWL:** Industry Foundation Classes Web Ontology Language (resource Description Framework/Web Ontology Language encoding of Industry Foundation Classes)
**KB:** knowledge base (ontology + instantiated data)
**KPI:** key performance indicator
**MQTT:** Message Queuing Telemetry Transport
**NTP:** Network Time Protocol
**OWL 2 DL:** Web Ontology Language 2 (description logic profile)
**OWL-Time:** World Wide Web Consortium vocabulary for temporal concepts
**PPE:** personal protective equipment
**PROV-O:** World Wide Web Consortium Provenance Ontology
**PTP:** Precision Time Protocol
**RDF:** Resource Description Framework
**RSP-QL:** Resource Description Framework Stream Processing-Query Language
**SCADA:** supervisory control and data acquisition
**SHACL:** Shapes Constraint Language
**SOSA:** sensor, observation, sample, and actuator
**SPARQL:** SPARQL Protocol and Resource Description Framework Query Language
**SSN:** Semantic Sensor Network Ontology
**SUS:** system usability scale
**SWRL:** Semantic Web Rule Language
**T-Box:** terminological box (schema/axioms of the ontology)
**UWB:** ultra-wideband
**VOC:** volatile organic compounds